

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A swimmer's score is given by $f(x) = x^2 + 2x - 1$ points for the 100m dash and $g(x) = 3x - 2$ points for the 200m dash. Find $f(g(x))$ and determine the domain of the composite function.

Q10. A tennis player's serve speed is given by $f(x) = 20 + 2x$ mph and the opponent's reaction time is given by $g(x) = 0.5x + 1$ seconds. Find $f(g(x))$ and interpret the result in the context of the game.

Chef's Tips

1

- Tip 1: Encourage students to simplify their expressions before evaluating.
- Tip 2: Remind students to consider the domain of each function when finding the composite.
- Tip 3: Suggest that students use real-world examples to help them understand the context of the problem.